

Supplementary notes - ITP-AL210926

I am really delighted you both enjoyed the course and it was useful in your learning journeys. Here are some notes that will help you in your self-study going forwards.

REPO: (you are here, but access will be revoked at end of next working week after completion!)

<https://github.com/lavenderlens/ITPP-AL210926>

VS Code shortcuts - Python and JS environments, Windows/MacOS

CTRL + B	toggle explorer pane / full screen
CTRL + +/-	up/down font size
CTRL + /	toggle comment on/off, any language
ALT click	multi-cursor insert
CTRL + D	multi-cursor match current selection
ALT + SHIFT + DOWN/UP	duplicate line down/up
=> WIN ALT	=> MAC OPT
=> WIN CTRL	=> MAC CMD

TOP LANGUAGES 2025

<https://www.geeksforgeeks.org/top-programming-languages/>

THE ASCII TABLE

<https://www.ascii-code.com/>

UML FOR SOFTWARE ENGINEERS

<https://www.geeksforgeeks.org/unified-modeling-language-uml-introduction/#3-types-of-uml-diagrams>

PSEUDOCODE

<https://www.geeksforgeeks.org/dsa/what-is-pseudocode-a-complete-tutorial/>

<https://www.geeksforgeeks.org/dsa/difference-between-algorithm-pseudocode-and-program/>

https://cs.wmich.edu/gupta/teaching/cs4310/lectureNotes_cs4310/PseudocodeBasics.pdf

Tooling for your own environment:

Download Python (**first**)

Then VS Code (**second**):

1. <https://www.python.org/downloads/>
2. <https://code.visualstudio.com/download>

Post-course self-study:

Independent tutorials:

Python

Jenkov - excellent tutorials.

<https://jenkov.com/tutorials/python/index.html>

RealPython - video courses paywalled but many articles free.

<https://realpython.com/>

W3: Basic but clear.

<https://www.w3schools.com/python/default.asp>

GeeksForGeeks

Intermediate, curated site of individual contributors, monetised from video courses sold same as RealPython, other content free.

<https://www.geeksforgeeks.org/python-programming-language-tutorial/>

OFFICIAL PYTHON

The last word: technical. Beware. Everything you seek is here, but for a major software vendor they aren't very user-friendly. I'd only ever go here if I couldn't find what I want through any of the other links above.

<https://docs.python.org/3/>

GENERAL

operator precedence including greater/less than

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Operators/Operator_precedence

INTRO: MIND CONTROLLED FLIGHT - I sometimes use this as a motivational intro as we spend the rest of the course typing stuff in and reading it out. This, which dates back to 2019 now, shows what is possible. It uses JavaScript libraries, but the principle is the same.

Interview with Charlie Gerard, mind controlled flight developer

<https://www.youtube.com/watch?v=Sp6egeaFNC4>

CUE TO 07:55

**PLAY FROM 07:55 - 10:05,
 17:40 - 20:20**

"Fail fast" - Facebook

Medium article

<https://medium.com/@devdevcharlie/experimenting-with-brain-computer-interfaces-in-javascript-8d6cb891fda8>

Demo

<https://www.youtube.com/watch?v=-PbpWYBRBgM>

mind control drone demo 15:30 - 17:05